

C-410A Vendor Performance Extract

Nordbrock RSV-160-V Rotary Screw Compressor - Field Performance Tables
Extract No. NB-RSV160-410A-2026-05 | Use for technician-level calculations and source reconciliation.

1. Resolved Model Curve: RSV-160-V, 250 kW motor, 460 V

Discharge pressure (psig)	Capacity at inlet conditions (acfm)	Shaft power (kW)	Package input current at 460 V (A)	Notes
90	1715	218	330	Factory test line; separator DP 6 psid
100	1660	231	350	Nominal
110	1585	245	371	Use for normal case interpolation
115	1540	253	383	High pressure band
120	1485	262	397	Near full-load current
125	1420	272	412	Avoid sustained operation without review

2. Correction Factors

Condition	Correction method	Reference value	Apply to
Ambient temperature	Corrected capacity = logged flow x (273.15 + ambient degC) / 298.15	25 degC baseline	Operating case airflow correction
Inlet filter differential pressure	Subtract 0.8% capacity for each psid above 0.5 psid	0.5 psid clean reference	Operating case airflow correction
Separator differential pressure	Flag service review when separator DP exceeds 10 psid	10 psid service review limit	Temperature and separator check
Receiver/filter pressure drop	Use measured or modelled downstream drop; do not treat package discharge pressure as end-use header pressure	instrument-specific	Pressure source reconciliation
Motor current	Estimated motor loading = motor current / confirmed FLA	392 A confirmed FLA	Power and loading check

3. Alarm and Maintenance Limits

Topic	Limit	Unit	Action wording
High discharge temperature alarm	230	degF	Investigate cooling, separator DP, lubricant condition, and control mode.
Separator differential pressure service limit	10	psid	Inspect separator element and oil return system.
Inlet filter differential pressure watch level	1.2	psid	Inspect inlet filter and intake ducting.
Motor current full-load reference	392	A	Use actual motor tag when available.
Motor bearing vibration review level	6.0	mm/s	Review mechanical condition; do not diagnose from a single reading.
Oil viscosity 40C low review level	42.0	cSt	Review lubricant condition and recent service record.

4. Notes for Field Use

- The capacity curve assumes clean inlet filter and separator differential pressure not exceeding 6 psid.
- When separator differential pressure exceeds 10 psid, use the curve for a baseline comparison only and flag condition as service-limited.
- The extract does not supersede plant permit or lockout procedures.
- Do not use package discharge pressure as end-use header pressure without adding receiver, filter, dryer, and piping corrections.

5. Example Interpolation Table

Pressure band	lower psig	upper psig	lower acfm	upper acfm	lower kW	upper kW
110 to 115 psig	110	115	1585	1540	245	253
115 to 120 psig	115	120	1540	1485	253	262
120 to 125 psig	120	125	1485	1420	262	272

End of vendor extract.